

Algebra II with Trigonometry

Chapter 6.1

Mental Math

Saved tutoring response with MathJax rendering. Original PDF file:
[4506904_alg-2trig-h-chp-6-1-note-docx.pdf](#)

Sure! Here are **10 mental math questions** based on the unit circle concepts from your PDF. These questions test core understanding and can be answered without writing anything down:

1.

What is the radius of the unit circle?

2.

What is the equation of the unit circle centered at the origin?

3.

If you start at (1,0) and move halfway around the unit circle, what are the coordinates of the point you reach?

4.

What is the length of the circumference of the unit circle?

5.

If $t = \pi/2$, what are the coordinates of the terminal point on the unit circle?

6.

If a terminal point lies in quadrant II, is its x-coordinate positive or negative?

7.

If $t = 2\pi$, where are you on the unit circle in relation to the starting point (1,0)?

8.

True or False: Adding 2π to an angle t on the unit circle changes the terminal point.

9.

If the y-coordinate of a point on the unit circle is positive, which quadrants could the point be in?

10.

If the x-coordinate of a point on the unit circle is 0, what are the possible values of y at this point?

Let me know if you want answers/explanations for each!